

PII DETECTION

2025-11-05 by Anh Vu

PURPOSE

The goal is to detect Personal Identifiable Information (PII) such as:

- Name
- Address
- Phone number
- SSN
- Email
- Credit card number

The above PII can be detected by regular expression (regex) but not completely. We need AI-assisted tools to enhance PII detection. Stanford CoreNLP has a Name Entity Recognition (NER) engine which can detect people's names and locations. This document shows what it can detect and its limitations.

OVERVIEW

Stanford CoreNLP is a Java-based natural language processing toolkit developed by Stanford NLP Group. It provides a suite of linguistic analysis tools, including Named Entity Recognition (NER) which identifies and classifies named entities in text into predefined categories. For PII detection, we leverage these entity types:

- PERSON: Names
- LOCATION: City, State, Country

CoreNLP works out-of-the-box. It can handle name variations like “Dr. Jon James”, “James, Jon”, and understand person names without keywords (context-aware).

It can detect city, state, and country where addresses usually have.

RESULTS

NAME

CoreNLP can detect person names out-of-the-box. Results are good.

Capability	Input	Result/Detection
First Last	Robin Williams	Robin Williams
First Middle Last	John Michael Fox	John Michael Fox
Name with Middle Initial	John Q. Public	John Q. Public
Name with Suffix	Michael Brown III	Michael Brown III (also work for Jr. Sr.)
Name with Title	Prof. Michael Douglas	Michael Douglas (also work for Mr. Ms. Dr.)
Name in a context	Patient Andy Lau was admitted on 2025-10-15	Andy Lau
International Name	Nguyễn Văn Thiệu Rajesh Kumar	Nguyễn Văn Thiệu Rajesh Kumar (also work for Spanish, French, German)
Only one First/Last name	Mary sent the email.	Mary (It can detect this but need more tests)
Avoid false positive	Send the report by Friday	No name detected, though Friday might be a person's name in some novels/movies.
Redaction	Andy Lau will meet with Mary Johnson tomorrow	Redacted text = [PERSON] will meet with [PERSON] tomorrow

ADDRESS

CoreNLP mainly detects locations such as city and country. If a given address includes a city and/or country, then it can detect the address. It cannot detect street addresses.

We use a regex to detect Zip codes here (hybrid approach). Out-of-the-box, it sometimes can detect a street name and/or state but not always. We need to actually train the model more to detect those entities better.

Capability	Input	Result/Detection
Full address	123 Main Street, Springfield, IL 62701	1 Locations found: Springfield 1 ZIP code found: 62701
Famous address (White House)	1600 Pennsylvania Avenue, Washington DC	Detected Pennsylvania Avenue and Washington DC. Sometimes it detects a street name or State abbreviation like DC, NY.
International address with country (UK Prime Minister)	10 Downing Street, London, UK	Detected London and UK. Same limitation with the street as above.
Street address only	123 Main Street	Not able to detect
Address in a context	Ship to 123 Main Street, Springfield, IL 62701 and bill to 456 Oak Avenue, New York, NY 10001.	Detected 3 locations: - Springfield - New York - NY Detected 2 ZIP codes: - 62701 - 10001
Redaction	Please send the package to Andy Lau at 123 Main Street, Springfield, IL 62701.	Redacted text = Please send the package to [PERSON] at 123 Main Street, [LOCATION], IL [ZIP_CODE].

PHONE NUMBER

We use regex to detect phone numbers. Currently, we can only detect US phone number formats, not international phone number formats. Our regex needs more work.

Capability	Input	Result/Detection
US format	(555) 123-4567	555) 123-4567
	555-123-4567	555-123-4567
	555.123.4567	555.123.4567
	5551234567	5551234567
	555-1234	Not detected
Invalid phone numbers	123456789012 and 12345	Not detected (correct)
Possible false positives	000-000-0000 111-111-1111	000-000-0000 111-111-1111 (incorrectly detect phone: False positive)
International formats	UK: +44 7911 123456 Germany: +49 30 12345678 France: +33 1 42 86 82 00 Japan: +81 3-1234-5678	Not detected

SOCIAL SECURITY NUMBER

We use regex to detect SSN. The result is not very good (contains false positives).

Capability	Input	Result/Detection
SSN regex detection	Employee records: 123-45-6789 hired on 2020-01-15 and 456-78-9012 hired on 02/20/2020	SSNs found: 2 - 123-45-6789 - 456-78-9012

EMAIL

We use regex to detect email.

Capability	Input	Result/Detection
EMAIL regex detection	Email me at john@example.com	john@example.com

CREDIT CARD NUMBER

We use the Luhn algorithm to detect credit card numbers. It calculates from right to left. So, the result might contain false positives.

Capability	Input	Result/Detection
VISA	4532015112830366	4532015112830366
Mastercard	5425233430109903	5425233430109903
Amex (15 digits)	374245455400126	374245455400126
Discover	6011000991300009	6011000991300009
Invalid CC number	1234-5678-9012-3456	no cc detected
Possible false positive	My account number is 1-4532-0151-1283-0366.	4532-0151-1283-0366 (detect as credit card)

COMPREHENSIVE TEST

The input is a mock form with all possible PII information. The output is a redacted version of the original.

Input

Patient Information: Full Name: Dr. John Michael Smith Jr. Also known as: Andy Hui, Lau Tak-wah Social Security Number: 123-45-6789 Contact Details: Primary Phone: (555) 123-4567 Secondary Phone: 555-987-6543

Mobile: +1-555-444-3333
Email: a.hui.lau@example.com
Alternate Email: alau99@healthcare.org

Address:

Home: 123 Main Street, Apt 4B, Springfield, IL 62701
Work: 456 Corporate Blvd, Suite 200, Chicago, IL 60601
Mailing: PO Box 9876, New York, NY 10001

Emergency Contact:

Name: Mary Elizabeth Johnson
Relationship: Spouse
Phone: 555.111.2222
Address: Same as patient

Payment Information:

Credit Card: 4532-0151-1283-0366 (Visa)
Cardholder: Andy Lau
Billing Address: 123 Main Street, Springfield, IL 62701

Additional Contacts:

Primary Physician: Dr. Sarah Williams, (555) 777-8888
Specialist: Robert Lee Johnson MD, 555-999-0000
Pharmacy: Springfield Pharmacy, 789 Oak Avenue, Springfield, IL 62701, (555) 222-3333

References:

Friend: Michael Davis, michael.davis@email.com, 555-444-5555
Colleague: Jennifer Brown at jennifer.b@company.com

Notes:

Patient relocated from Los Angeles, CA to Springfield, IL.
Previous SSN on file: 234-56-7890 (update required)
Former address: 999 Sunset Boulevard, Los Angeles, CA 90001

Output

Patient Information:

Full Name: Dr. [PERSON] Jr.
Also known as: [PERSON], [PERSON] Tak-wah
Social Security Number: [SSN]

Contact Details:

Primary Phone: ([PHONE])
Secondary Phone: [PHONE]

Mobile: +[PHONE]

Email: [EMAIL]

Alternate Email: [EMAIL]

Address:

Home: 123 Main Street, Apt 4B, [LOCATION], IL [ZIP_CODE]

Work: 456 Corporate Blvd, Suite 200, [LOCATION], IL [ZIP_CODE]

Mailing: PO Box 9876, [LOCATION], [LOCATION] [ZIP_CODE]

Emergency Contact:

Name: [PERSON]

Relationship: Spouse

Phone: [PHONE]

Address: Same as patient

Payment Information:

Credit Card: [CREDIT_CARD] (Visa)

Cardholder: [PERSON]

Billing Address: 123 Main Street, [LOCATION], IL [ZIP_CODE]

Additional Contacts:

Primary Physician: Dr. [PERSON], ([PHONE]

Specialist: [PERSON] MD, [PHONE]

Pharmacy: [ORGANIZATION], 789 Oak Avenue, [LOCATION], IL [ZIP_CODE], ([PHONE]

References:

Friend: [PERSON], [EMAIL], [PHONE]

Colleague: [PERSON] at [EMAIL]

Notes:

Patient relocated from [LOCATION], CA to [LOCATION], IL.

Previous SSN on file: [SSN] (update required)

Former address: 999 [LOCATION], [LOCATION], CA [ZIP_CODE]